

Hackathon Task: Intelligent Face Tracker with Auto-Registration and Visitor Counting

1. Project Planning

Section	Contents
1. Project Planning	Problem statement + 5-phase planning breakdown (Architecture → Core → Logging → Dashboard → Optimization)
2. Feature Documentation	All 12 features documented with parameters, defaults, and behavior (detection, recognition, unique counting, tracking, pacing, upload, RTSP, dashboard, DB, etc.)
3. Compute Load Estimation	CPU vs. GPU load tables, time-per-frame breakdowns, memory usage, and hardware tier recommendations
4. AI Planning Document	Full system architecture diagram, key AI design decisions with rationale, and re-identification failure modes + mitigations
5. Setup Instructions	Virtual env setup, pip install, GPU support, model files verification, 3 run options (dashboard / CLI / utilities)
6. Assumptions Made	10 documented assumptions (camera angle, lighting, concurrency, SQLite, FPS metadata, etc.)
7. Sample config.json	Full sample JSON + complete parameter reference table with types, defaults, and descriptions
8. Directory Structure	Annotated full project tree
9. Tech Stack	Complete technology table across all 14 modules

2.Feature Documentation

2.1 Face Detection

1. Model: YOLOv8-nano-face (yolov8n-face.pt)
2. Inference size: 640×640 input (auto-resized)
3. Configurable confidence threshold (confidence_threshold, default: 0.45)
4. Minimum face size filter (min_face_size, default: 25px) eliminates tiny false positives
5. GPU acceleration: Automatically uses CUDA if NVIDIA GPU is available; falls back to CPU
6. Output: Bounding boxes (x1, y1, x2, y2) + confidence score + cropped face ROI

2.2 Face Recognition & Auto-Registration

1. Model: ArcFace InsightFace MobileFaceNet (w600k_mbf.onnx)
2. Embedding dimensionality: 512-dimensional L2-normalized facial feature vector
3. Similarity metric: Cosine Similarity between unit embeddings
4. Threshold: (similarity_threshold, default: 0.30)
5. Auto-Registration: New faces automatically assigned unique sequential IDs (VISITOR_0001, VISITOR_0002, ...)
6. Adaptive Template Refinement: On each successful re-identification, stored embedding is updated via exponential moving average ($0.85 \times \text{old} + 0.15 \times \text{new}$) to adapt to lighting/pose drift over time

2.3 Unique Visitor Counting (Core Feature)

1. Maintained via `SELECT COUNT(*) FROM visitors` — one row per unique face
2. Re-identification of returning visitors never increments the unique count
3. Persistent across multiple sessions and stream restarts
4. Retrievable via Dashboard API: `GET http://localhost:8080/api/stats`
5. CLI: `python main.py --stats`
6. DB Inspector: `python check_db.py`
7. Raw SQL: `SELECT COUNT(*) FROM visitors;`

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2.4 Multi-Object Tracking (IoU-Based)

1. Algorithm: Greedy IoU cost matrix matching
2. IoU Threshold: (iou_match_threshold, default: 0.35)
3. Frame Skipping: (detection_skip_frames, default: 2) — full YOLO detection runs every N+1 frames
4. Max Disappeared Frames: (max_disappeared_frames, default: 25) — tracks persist through brief occlusions before exit is logged

2.5 Entry & Exit Event Logging

1. Entry Event: Logged when track accumulates ≥ 2 detection hits
2. Exit Event: Logged when track disappears for more than max_disappeared_frames
3. Each event records: visitor_id, event_type, timestamp, confidence, image_path
4. Face crops stored under logs/entries/YYYY-MM-DD/ or logs/exits/YYYY-MM-DD/

2.6 Real-Time Frame Pacing (Anti-Lag)

1. Reads native video FPS via cv2.CAP_PROP_FPS
2. Wall-clock synchronization prevents slow-motion playback
3. Dynamically drops intermediate frames when processing falls behind

2.7 Video File Upload

1. Browser drag-and-drop or file picker for MP4 / AVI / MKV / MOV files
2. Uploaded files saved to data/uploads/
3. REST endpoint: POST http://localhost:8080/api/upload_video

2.8 Live RTSP Camera Support

1. Connects to live RTSP cameras (rtsp://username:password@ip:port/stream)
2. Automatic reconnection with configurable rtsp_reconnect_delay

2.9 Web Dashboard (AetherVision UI)

1. Tab 1 — Live Vision Feed: MJPEG stream, HUD overlay, pipeline controls, upload, calibration sliders
2. Tab 2 — Analytics Hub: KPI cards, hourly footfall bar chart, entry/exit donut chart

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3. Tab 3 — Visitors Gallery: All registered profiles with face thumbnails, visit counts, alias assignment
4. Tab 4 — Audit Trail: Full event table with face snapshots, confidence scores, CSV export
5. Tab 5 — System Console: Live events.log terminal, database reset

2.10 Database Persistence

1. SQLite with WAL journal mode for concurrent read/write
2. visitors table: Unique profiles + 512-dim embedding BLOB
3. events table: All entry/exit events with full metadata

2.11 Annotated Video Output

1. Saves full annotated MP4 with bounding boxes, visitor IDs, and HUD overlay

2.12 Configuration & CLI

1. All parameters managed via config.json
2. CLI options: --source, --skip, --conf, --sim, --no-display, --stats, --reset-db.

3. Compute Load Estimation

3.1 CPU-Only Mode (Default)

Component	Estimated CPU Load	Time per Frame
YOLOv8-nano Detection	~60–80% single core	80–150 ms
ArcFace Embedding Extraction	~40–60% single core	30–60 ms
IoU Tracking Matrix	~5%	< 2 ms
SQLite Read/Write	~5%	2–8 ms
MJPEG JPEG Encoding	~10%	5–15 ms
Total (full detection frame)	~80–120% CPU	~120–230 ms
Effective FPS (skip_frames=2)	—	~15–25 FPS

Detection runs every 3rd frame. Skip frames cost only IoU tracking (~2 ms), maintaining 25–35 FPS display throughput.

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3.2 GPU Mode (NVIDIA CUDA)

Component	GPU Load	Time per Frame
YOLOv8-nano Detection	~30–50% GPU	8–20 ms
ArcFace Embedding Extraction	~20–40% GPU	5–15 ms
IoU Tracking + DB	~5% CPU	3–10 ms
Total (full detection frame)	~40–70% GPU	~15–45 ms
Effective FPS (skip_frames=2)	—	~40–70 FPS

3.3 Memory Usage

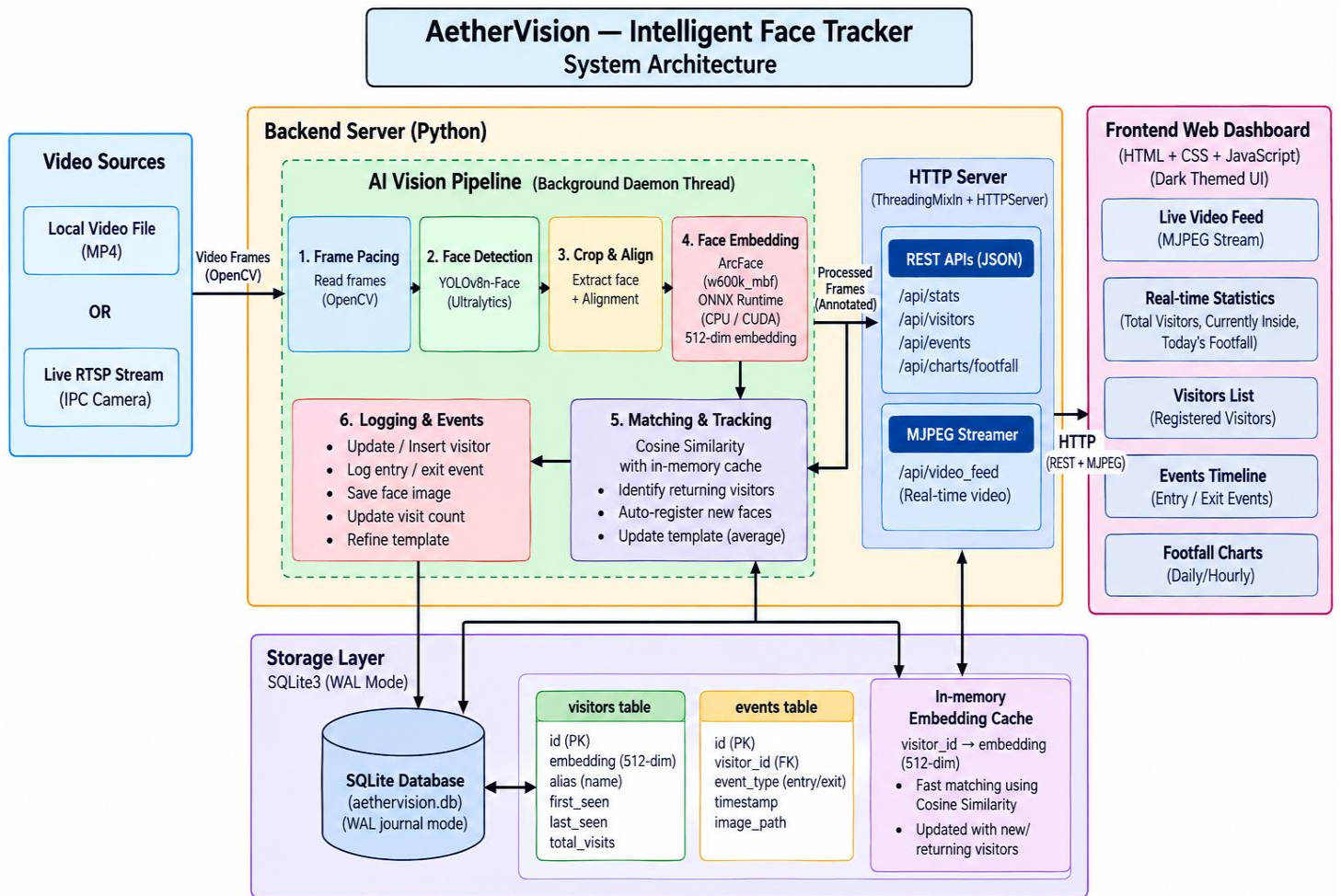
Resource	CPU Mode	GPU Mode
RAM	400–700 MB	400–700 MB
VRAM	—	300–500 MB
SQLite DB (1000 visitors)	~8–12 MB	~8–12 MB

3.4 Recommended Hardware

Tier	Hardware	Expected FPS
Minimum	Intel Core i5 / 8GB RAM	10–20 FPS
Recommended CPU	Intel Core i7/i9, 16GB RAM	20–30 FPS
Optimal (GPU)	NVIDIA RTX 3060+, 16GB RAM	40–70 FPS

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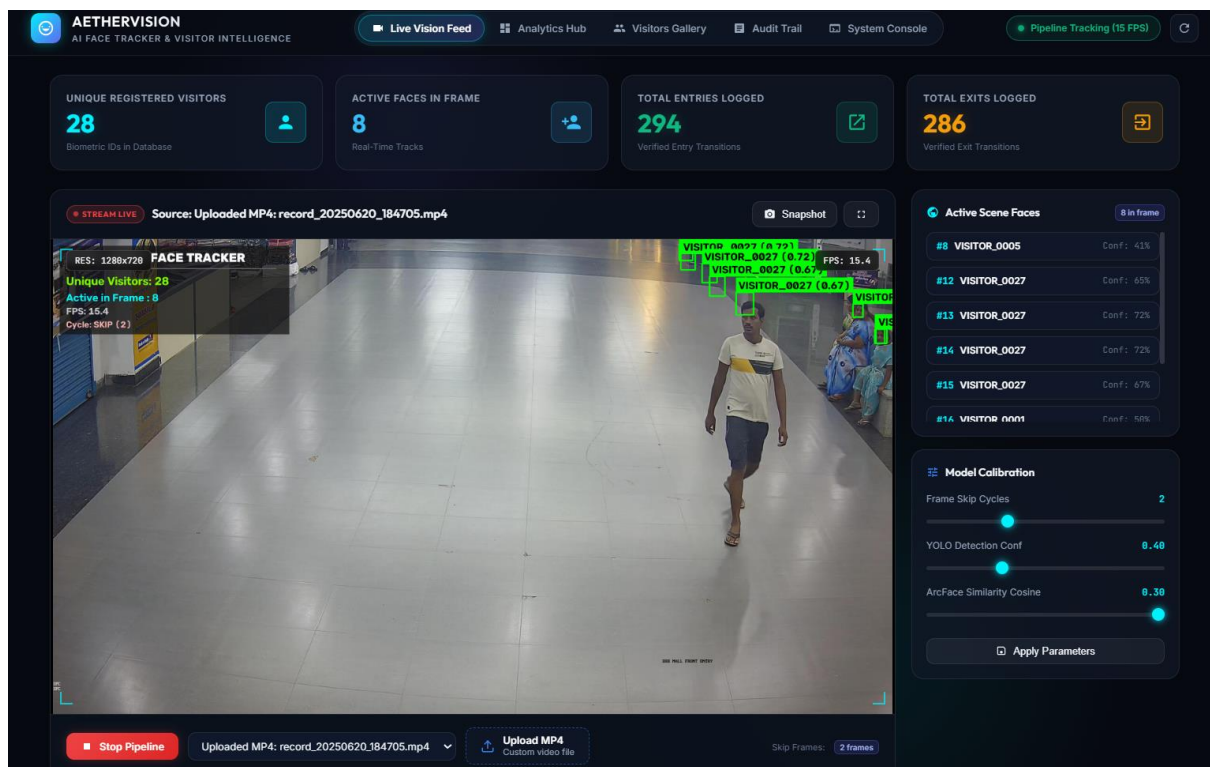
Architecture diagram



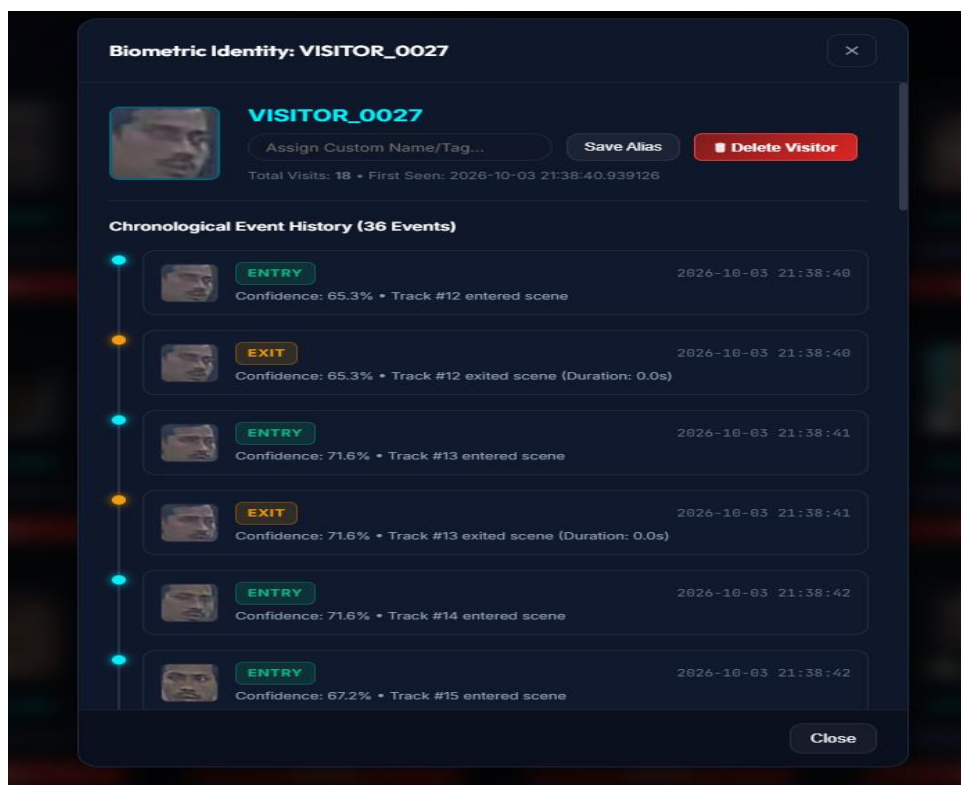
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Sample Output:

Identifying the Visitor:



Entry and exit of the Visitor :



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Database of all Visitors:

Database Structure Browse Data Edit Pragmas Execute SQL						
Table: events						
	visitor_id	event_type	timestamp	image_path	confidence	details
	Filter	Filter	Filter	Filter	Filter	Filter
1	VISITOR_0001	entry	2026-10-03 20:48:50	logs\entries\2026-10-03\VISITOR_0001...	0.435463517904282	Track #1 entered scene
2	VISITOR_0002	entry	2026-10-03 20:48:55	logs\entries\2026-10-03\VISITOR_0002...	0.401861608028412	Track #2 entered scene
3	VISITOR_0001	entry	2026-10-03 20:48:56	logs\entries\2026-10-03\VISITOR_0001...	0.403250306844711	Track #3 entered scene
4	VISITOR_0002	entry	2026-10-03 20:48:57	logs\entries\2026-10-03\VISITOR_0002...	0.427103966474533	Track #4 entered scene
5	VISITOR_0002	entry	2026-10-03 20:48:58	logs\entries\2026-10-03\VISITOR_0002...	0.436153173446655	Track #5 entered scene
6	VISITOR_0001	exit	2026-10-03 20:48:53	logs\exits\2026-10-03\VISITOR_0001_e...	0.429187297821045	Track #1 exited scene (Duration: ...
7	VISITOR_0003	entry	2026-10-03 20:48:59	logs\entries\2026-10-03\VISITOR_0003...	0.657139182090759	Track #6 entered scene
8	VISITOR_0003	entry	2026-10-03 20:49:00	logs\entries\2026-10-03\VISITOR_0003...	0.467280268669128	Track #7 entered scene
9	VISITOR_0002	exit	2026-10-03 20:48:55	logs\exits\2026-10-03\VISITOR_0002_e...	0.401861608028412	Track #2 exited scene (Duration: ...
10	VISITOR_0003	entry	2026-10-03 20:49:01	logs\entries\2026-10-03\VISITOR_0003...	0.45873299241066	Track #8 entered scene
11	VISITOR_0003	entry	2026-10-03 20:49:01	logs\entries\2026-10-03\VISITOR_0003...	0.502240061759949	Track #9 entered scene
12	VISITOR_0001	exit	2026-10-03 20:48:56	logs\exits\2026-10-03\VISITOR_0001_e...	0.403250306844711	Track #3 exited scene (Duration: ...
13	VISITOR_0003	entry	2026-10-03 20:49:02	logs\entries\2026-10-03\VISITOR_0003...	0.69777649641037	Track #10 entered scene
14	VISITOR_0002	exit	2026-10-03 20:48:57	logs\exits\2026-10-03\VISITOR_0002_e...	0.427103966474533	Track #4 exited scene (Duration: ...
15	VISITOR_0003	entry	2026-10-03 20:49:02	logs\entries\2026-10-03\VISITOR_0003...	0.683742523193359	Track #11 entered scene
16	VISITOR_0002	exit	2026-10-03 20:48:58	logs\exits\2026-10-03\VISITOR_0002_e...	0.436153173446655	Track #5 exited scene (Duration: ...
17	VISITOR_0003	entry	2026-10-03 20:49:04	logs\entries\2026-10-03\VISITOR_0003...	0.676365673542023	Track #12 entered scene
18	VISITOR_0003	entry	2026-10-03 20:49:04	logs\entries\2026-10-03\VISITOR_0003...	0.716625392436981	Track #13 entered scene
19	VISITOR_0004	entry	2026-10-03 20:49:04	logs\entries\2026-10-03\VISITOR_0004...	0.400768876075745	Track #14 entered scene
20	VISITOR_0003	entry	2026-10-03 20:49:05	logs\entries\2026-10-03\VISITOR_0003...	0.721234321594238	Track #15 entered scene
21	VISITOR_0003	exit	2026-10-03 20:48:59	logs\exits\2026-10-03\VISITOR_0003_e...	0.657139182090759	Track #6 exited scene (Duration: ...
22	VISITOR_0003	entry	2026-10-03 20:49:06	logs\entries\2026-10-03\VISITOR_0003...	0.713218927383423	Track #16 entered scene
23	VISITOR_0003	exit	2026-10-03 20:49:00	logs\exits\2026-10-03\VISITOR_0003_e...	0.467280268669128	Track #7 exited scene (Duration: ...
24	VISITOR_0003	entry	2026-10-03 20:49:06	logs\entries\2026-10-03\VISITOR_0003...	0.716456472873688	Track #17 entered scene
25	VISITOR_0003	exit	2026-10-03 20:49:01	logs\exits\2026-10-03\VISITOR_0003_e...	0.45873299241066	Track #8 exited scene (Duration: ...

Entry & Exit of Visitor:

Visitor-0027:

DB Browser for SQLite - D:\Intelligent Face Tracker\data\visitor_system.db						
File Edit View Tools Help						
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database						
Database Structure Browse Data Edit Pragmas Execute SQL						
Table: events						
	visitor_id	event_type	timestamp	image_path	confidence	details
	Filter	Filter	Filter	Filter	Filter	Filter
1	VISITOR_0027	entry	2026-10-03 21:38:40	logs\entries\2026-10-03\VISITOR_0027...	0.653356730937958	Track #12 entered scene
2	VISITOR_0027	entry	2026-10-03 21:38:41	logs\entries\2026-10-03\VISITOR_0027...	0.715940058231354	Track #13 entered scene
3	VISITOR_0027	entry	2026-10-03 21:38:42	logs\entries\2026-10-03\VISITOR_0027...	0.716381728649139	Track #14 entered scene
4	VISITOR_0027	entry	2026-10-03 21:38:42	logs\entries\2026-10-03\VISITOR_0027...	0.671710669994354	Track #15 entered scene
5	VISITOR_0027	entry	2026-10-03 21:38:43	logs\entries\2026-10-03\VISITOR_0027...	0.666122853755951	Track #18 entered scene
6	VISITOR_0027	entry	2026-10-03 21:38:44	logs\entries\2026-10-03\VISITOR_0027...	0.68600058555603	Track #19 entered scene
7	VISITOR_0027	entry	2026-10-03 21:38:44	logs\entries\2026-10-03\VISITOR_0027...	0.749910414218903	Track #20 entered scene
8	VISITOR_0027	entry	2026-10-03 21:38:45	logs\entries\2026-10-03\VISITOR_0027...	0.750924110412598	Track #21 entered scene
9	VISITOR_0027	entry	2026-10-03 21:38:45	logs\entries\2026-10-03\VISITOR_0027...	0.769296705722809	Track #22 entered scene
10	VISITOR_0027	exit	2026-10-03 21:38:40	logs\exits\2026-10-03\VISITOR_0027_e...	0.653356730937958	Track #12 exited scene (Duration: ...
11	VISITOR_0027	exit	2026-10-03 21:38:41	logs\exits\2026-10-03\VISITOR_0027_e...	0.715940058231354	Track #13 exited scene (Duration: ...
12	VISITOR_0027	exit	2026-10-03 21:38:42	logs\exits\2026-10-03\VISITOR_0027_e...	0.716381728649139	Track #14 exited scene (Duration: ...
13	VISITOR_0027	exit	2026-10-03 21:38:42	logs\exits\2026-10-03\VISITOR_0027_e...	0.671710669994354	Track #15 exited scene (Duration: ...
14	VISITOR_0027	exit	2026-10-03 21:38:43	logs\exits\2026-10-03\VISITOR_0027_e...	0.666122853755951	Track #18 exited scene (Duration: ...
15	VISITOR_0027	exit	2026-10-03 21:38:44	logs\exits\2026-10-03\VISITOR_0027_e...	0.68600058555603	Track #19 exited scene (Duration: ...
16	VISITOR_0027	exit	2026-10-03 21:38:44	logs\exits\2026-10-03\VISITOR_0027_e...	0.749910414218903	Track #20 exited scene (Duration: ...
17	VISITOR_0027	exit	2026-10-03 21:38:45	logs\exits\2026-10-03\VISITOR_0027_e...	0.750924110412598	Track #21 exited scene (Duration: ...
18	VISITOR_0027	exit	2026-10-03 21:38:45	logs\exits\2026-10-03\VISITOR_0027_e...	0.769296705722809	Track #22 exited scene (Duration: ...
19	VISITOR_0027	entry	2026-10-03 23:58:35	logs\entries\2026-10-03\VISITOR_0027...	0.653356730937958	Track #12 entered scene
20	VISITOR_0027	entry	2026-10-03 23:58:36	logs\entries\2026-10-03\VISITOR_0027...	0.715940058231354	Track #13 entered scene
21	VISITOR_0027	entry	2026-10-03 23:58:37	logs\entries\2026-10-03\VISITOR_0027...	0.716381728649139	Track #14 entered scene
22	VISITOR_0027	entry	2026-10-03 23:58:38	logs\entries\2026-10-03\VISITOR_0027...	0.671710669994354	Track #15 entered scene
23	VISITOR_0027	entry	2026-10-03 23:58:40	logs\entries\2026-10-03\VISITOR_0027...	0.666122853755951	Track #18 entered scene
24	VISITOR_0027	entry	2026-10-03 23:58:42	logs\entries\2026-10-03\VISITOR_0027...	0.68600058555603	Track #19 entered scene
25	VISITOR_0027	entry	2026-10-03 23:58:43	logs\entries\2026-10-03\VISITOR_0027...	0.749910414218903	Track #20 entered scene

Logs:

1. System Events Log :

The primary plain-text log file that records the underlying operations of the AI pipeline.

- Initialization & State: Logs pipeline startups, model initializations, and graceful shutdowns.
- Biometric Performance: Records the latency (in milliseconds) required to generate the 512-dimensional facial embeddings for profiling.
- Recognition Confidence: Logs whenever a live track successfully matches an existing biometric template, including the specific cosine similarity score.
- Exception Handling: Captures dropped frames, unhandled connection aborts, and database locks for debugging purposes.

2. Visual Audit Snapshots :

To provide undeniable proof of events, the system logs cropped facial snapshots rather than relying strictly on text.

- Entry Snapshots: When an individual enters the camera frame and meets the minimum tracking threshold, a localized face thumbnail is saved to a timestamped folder in logs/entries/.
- Exit Snapshots: When the tracking algorithm determines a face has departed the frame (exceeding the max_disappeared_frames threshold), a final reference snapshot is written to logs/exits/.

3. Database Event Ledger :

The SQLite database maintains an immutable events table that acts as the backbone for the web dashboard's Audit Trail. Every physical transition creates a structured row containing:

- The assigned visitor_id (e.g., VISITOR_0027)
- The precise ISO-formatted timestamp of the event.
- The event classification (entry or exit).
- The relative filesystem path mapping directly to the corresponding visual audit snapshot.

4. Annotated Video Archiving :

Controlled by the save_annotated_video flag in config.json, the system can optionally render and export a permanent video record. This output file bakes the YOLO bounding boxes, assigned Track IDs, and real-time model confidence percentages directly into the video frames, providing a complete historical recreation of the session.